AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 10 · WORKBOOK

If/Then Decisions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected conditional trace.

Trace updates, compare boundary language, reject an extra action and handle unavailable input.

Keep one habit: Keep the condition result and branch evidence with the final answer.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Algorithms as Recipes

Categories and Labels

Rules or Learned Patterns?

W1 · Trace the paper game

> Show the update before the condition and the final branch result.

Your evidence record

- Use the paper-game procedure: add 2 to score; if score is at least 6 subtract 3, otherwise add 4; report and stop. Start at 2, then make a separate run starting at 5.

  For each run record the value at the **condition**, its true or false result, the **branch** and the final score.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Read four boundary phrases

> Say which comparisons include equality.

Your evidence record

- The known value at a **condition** is exactly 6. Decide whether each test is true or false: at least 6; above 6; at most 6; below 6. Explain which phrases include equality.

  Then name one value below and one value above 6 for neighbouring tests.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Reject an invented extra branch

> Trace the one decision and its report-and-stop instruction.

Your evidence record

- In a separate decision the current score is 4. If score is at least 5 subtract 2; otherwise add 3; report and stop. A learner adds 3 to get 7, then subtracts 2 because 7 is now at least 5, and reports 5.

  Identify the extra action and give the correct trace and result.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Handle an unknown score

> Name the missing evidence and request it before choosing.

Your evidence record

- The score is unreadable at the at-least-6 test. A helper says the test must be false and takes the otherwise **branch**. Explain what is unknown and why that choice is unjustified.

  State the next step required by this paper exercise and give two possible scores that would select different **branch**es.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Transfer the method to a new display

> Keep one first attempt before hints or feedback.

Read and explain independently

A paper brightness display adds 3 to its starting number. If the current brightness is at most 7, subtract 2; otherwise add 1. Report and stop. Trace separate runs from starting values 3, 4 and 5.

Identify the run that tests equality. Then show how its result changes if at most 7 is accidentally written as below 7.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Transfer the method to a new display

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Add one transparent if/then **branch** and one **else** path to the Classroom Sorter.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Read four boundary phrases for **Condition**s and boundary cases. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Condition**s and boundary cases. Proposed explanation: “The **condition** reads the starting value”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A club admits a visitor if they have a ticket AND an accompanying adult. Test all four yes/no combinations. A student changes AND to OR. Which outcomes change?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

If a card has a star, send it to A; otherwise send it to B.

Sort a star, a circle, and a card with both shapes.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Use this exact rule: if a card has more than three dots, place it in tray A; otherwise use tray B. Trace cards with 2, 3 and 4 dots.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 10 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A club admits a visitor if they have a ticket AND an accompanying adult. Test all four yes/no combinations. A student changes AND to OR. Which outcomes change?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A paper brightness display adds 3 to its starting number. If the current brightness is at most 7, subtract 2; otherwise add 1. Report and stop. Trace separate runs from starting values 3, 4 and 5.

Identify the run that tests equality. Then show how its result changes if at most 7 is accidentally written as below 7.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

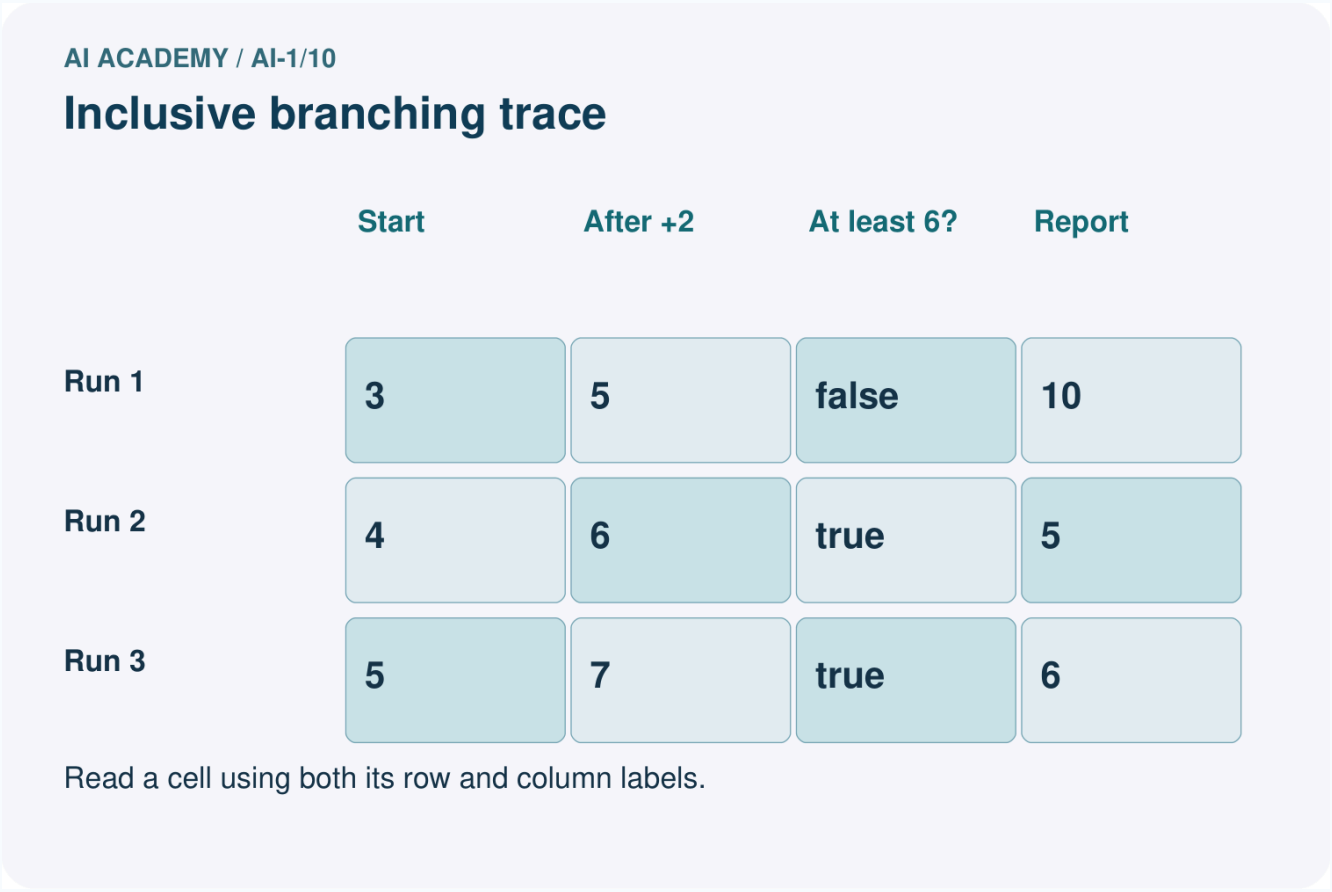
> Keep the explanation and the support you used.

Use this when helpful

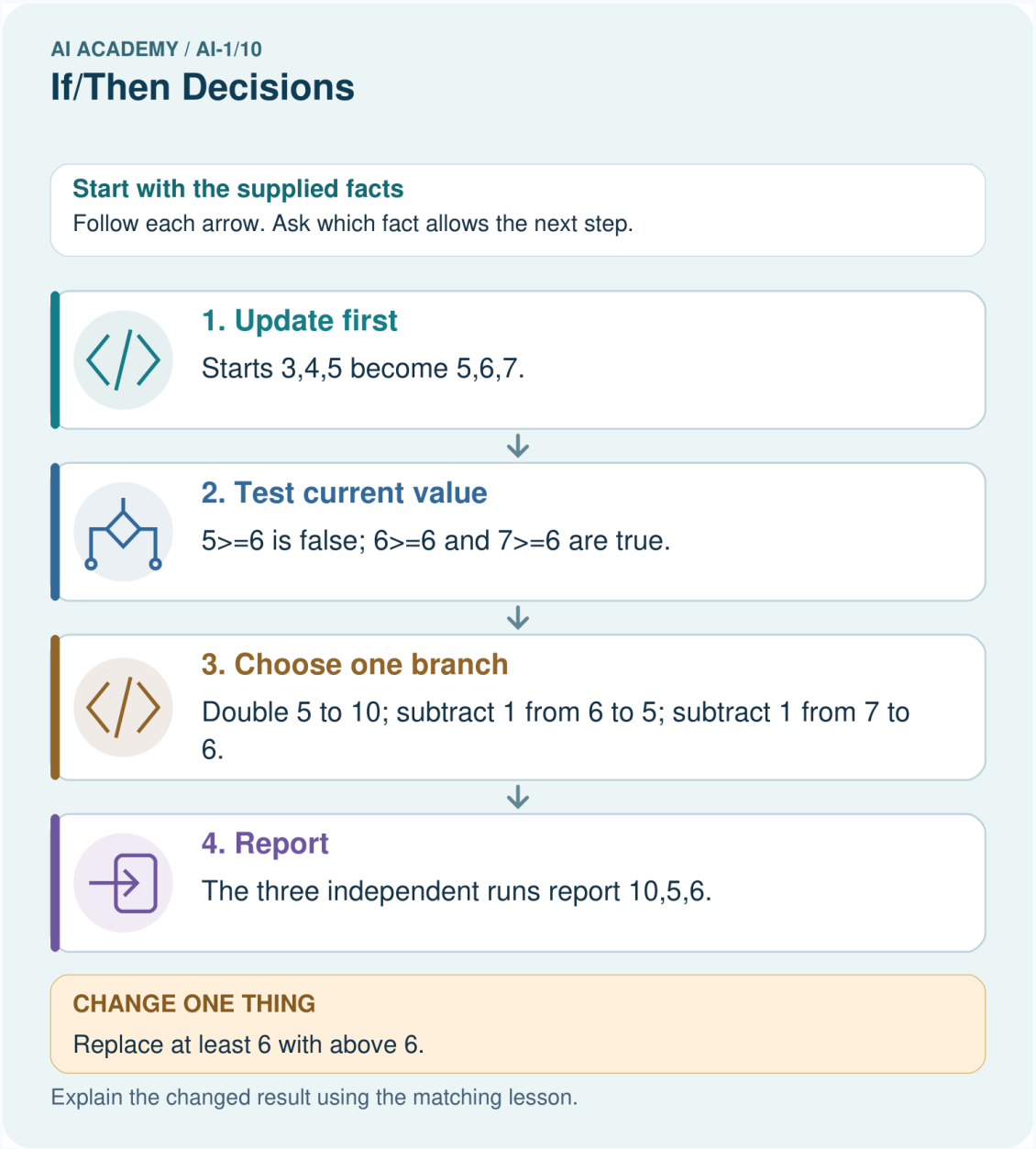
A fictional counter begins with a number, adds 2, then chooses one **branch**.

If the updated value is at least 6, subtract 1; otherwise double it.

Report once and stop. Use separate starting values 3,4,5.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Which **input** detects the difference between the two written **condition**s? Explain its entire run.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Include equality examples when reviewing sorter **condition**s, and trace from original **input** through **output**.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Include equality examples when reviewing sorter **condition**s, and trace from original **input** through **output**.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional counter begins with a number, adds 2, then chooses one **branch**.
- Which labelled step matches this detail? The three independent runs report 10,5,6.
- 1. Update first
- 2. Report
- 3. Choose one **branch**
- 4. Test current value
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Report The three independent runs report 10,5,6. The **condition** is evaluated after adding 2.

Equality belongs to the at-least **branch**, so starting at 4 checks the boundary. Once a **branch** finishes, the program reports; it does not reconsider the **condition** just because doubling 5 produced 10.

Changed-case question

- Changed-case practice: Replace at least 6 with above 6. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Only the run starting at 4 changes: it reports 12.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed **condition** and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Only the run starting at 4 changes: it reports 12. Updated value 6 fails a strict above comparison and is doubled.

Repair and save your work

Review Algorithms as Recipes (ai-1/9). Goal: Trace an ordered algorithm using the current value after every instruction. Short practice: Use the arrows to explain where the two zero-start runs first differ..

Return to this week and explain what you changed.

Review Categories and Labels (ai-1/7). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Explain why B originally was revise rather than review, then label its changed version..

Return to this week and explain what you changed.

Review Rules or Learned Patterns? (ai-1/4). Goal: Execute a stated rule including its boundary. Short practice: Show which stage used labels and which stage used only the new measurement..

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. If/Then Decisions: the case

AI ACADEMY / AI-1/10 · 01 · LOOK AT THE FACTS

If/Then Decisions: the case

The situation

A fictional counter begins with a number, adds 2, then chooses one branch.

Supplied evidence

If the updated value is at least 6, subtract 1; otherwise double it.
Report once and stop. Use separate starting values 3,4,5.

By the end, I can trace exactly one branch and test a boundary that includes equality.

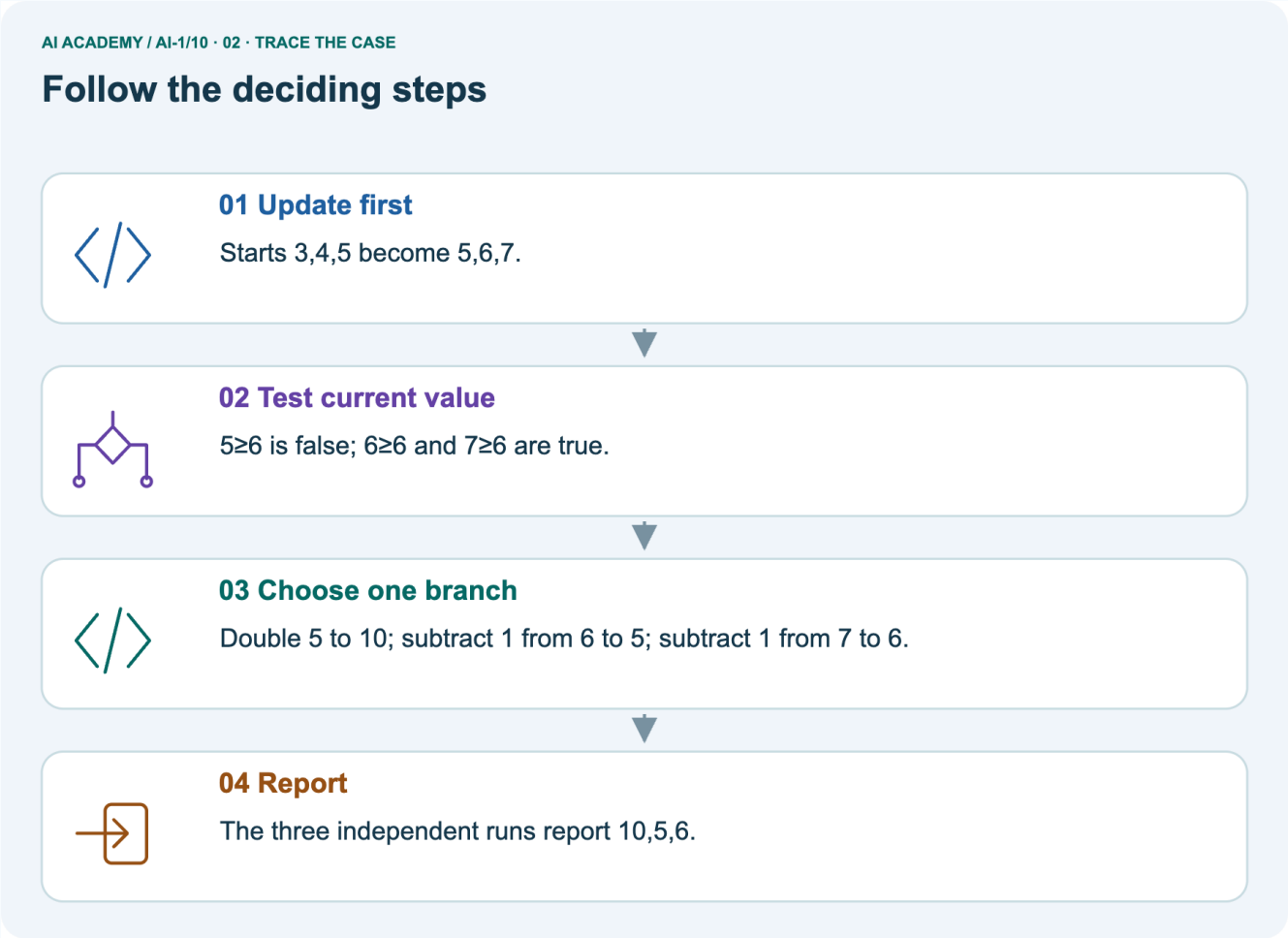
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional counter begins with a number, adds 2, then chooses one **branch**. Supplied facts: If the updated value is at least 6, subtract 1; otherwise double it. Report once and stop. Use separate starting values 3,4,5.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



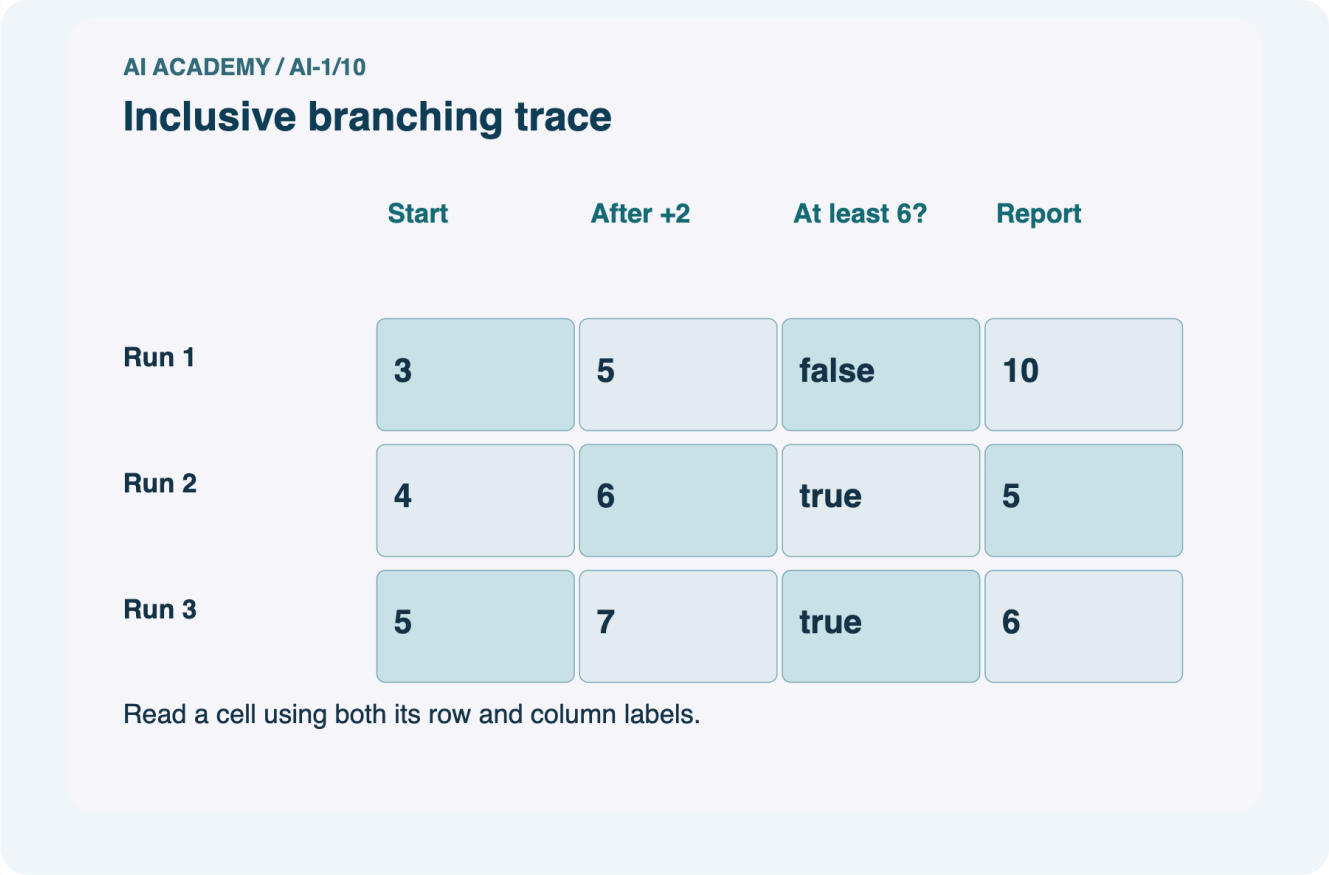
Follow the labelled picture with the matching explanation. Open the vector diagram

Update first: Starts 3,4,5 become 5,6,7. → Test current value: 5≥6 is false; 6≥6 and 7≥6 are true. → Choose one **branch**: Double 5 to 10; subtract 1 from 6 to 5; subtract 1 from 7 to 6. → Report: The three independent runs report 10,5,6.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Inclusive branching trace



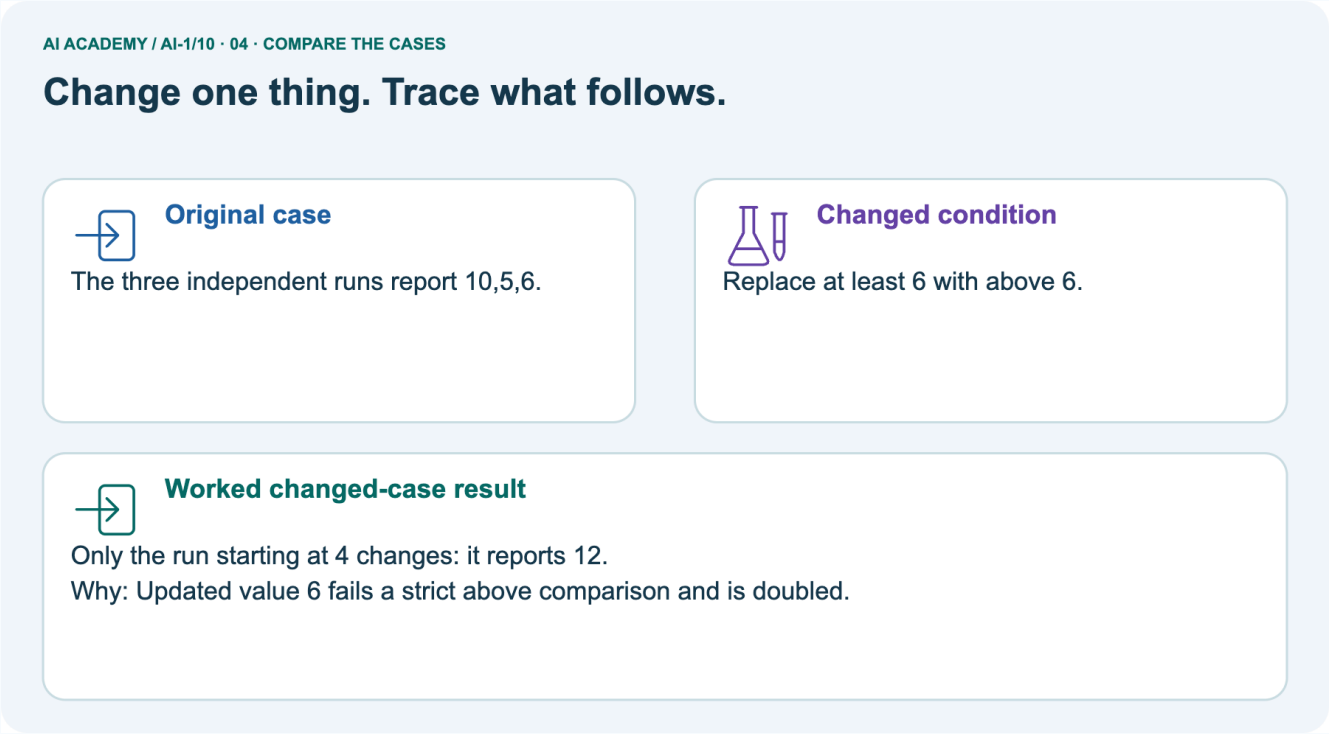
Follow the labelled picture with the matching explanation. Open the vector diagram

Inclusive **branch**ing trace. Run 1: Start = 3, After +2 = 5, At least 6? = false, Report = 10. Run 2: Start = 4, After +2 = 6, At least 6? = true, Report = 5. Run 3: Start = 5, After +2 = 7, At least 6? = true, Report = 6.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

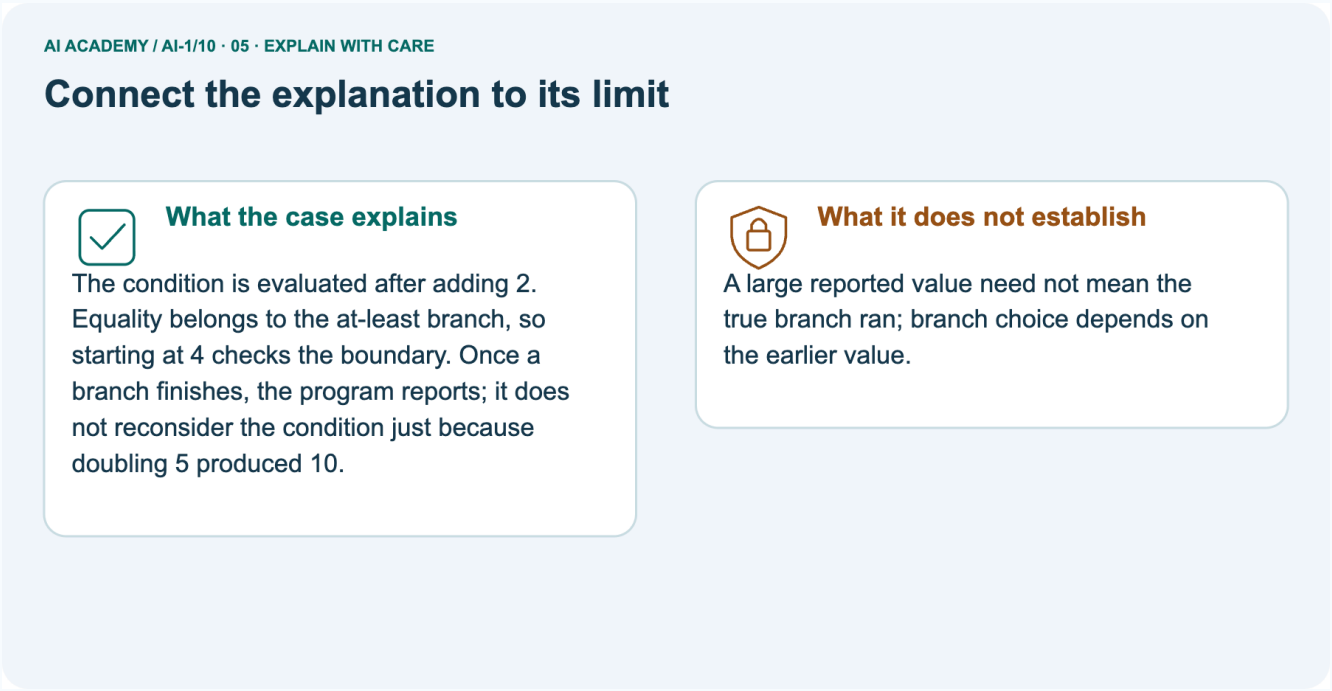
Original case: The three independent runs report 10,5,6. Changed **condition**: Replace at least 6 with above 6. Expected result: Only the run starting at 4 changes: it reports 12.

Reason: Updated value 6 fails a strict above comparison and is doubled.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The **condition** is evaluated after adding 2. Equality belongs to the at-least **branch**, so starting at 4 checks the boundary.

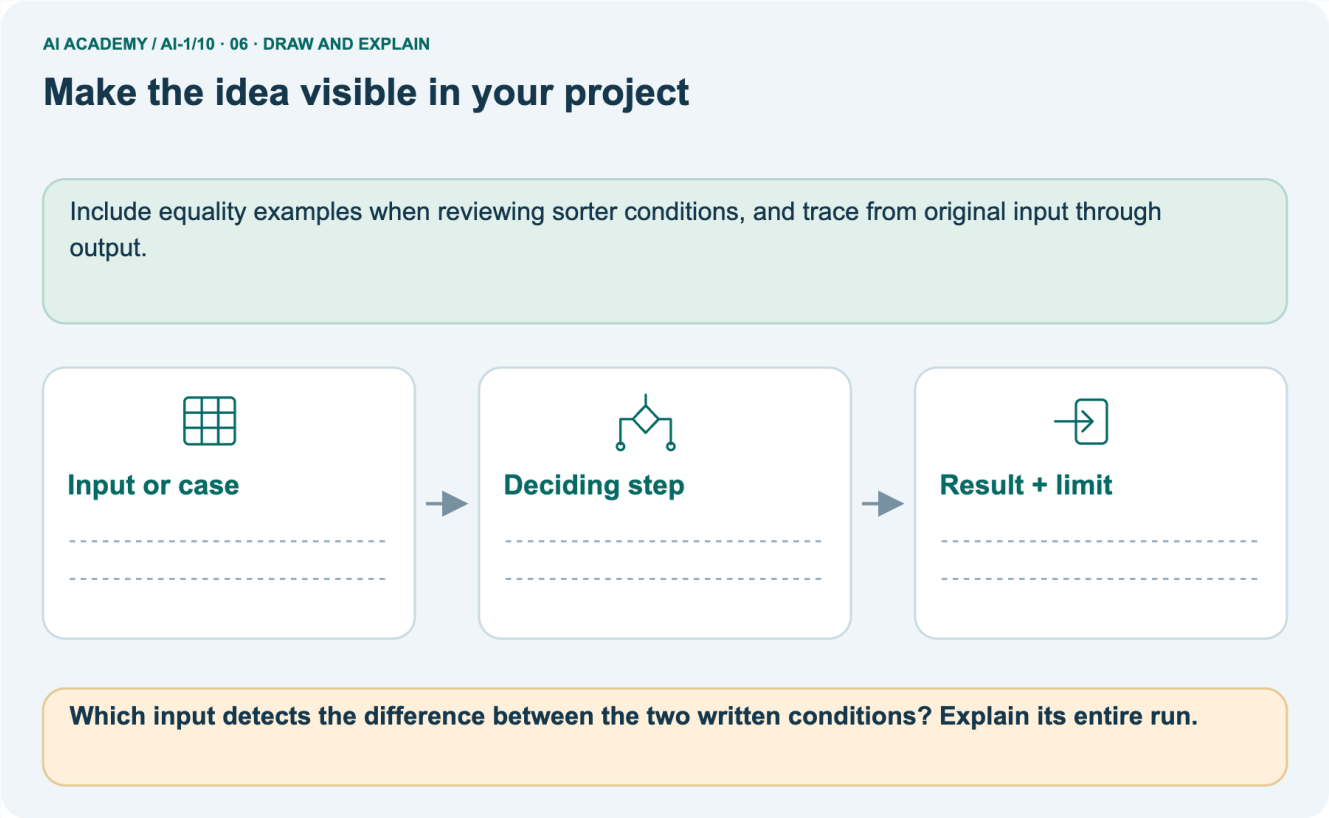
Once a **branch** finishes, the program reports; it does not reconsider the **condition** just because doubling 5 produced 10.

Limit: A large reported value need not mean the true **branch** ran; **branch** choice depends on the earlier value.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Include equality examples when reviewing sorter **condition**s, and trace from original **input** through **output**. Draw your **input**, deciding step, result and limit.

Supported question: Which **input** detects the difference between the two written **condition**s? Explain its entire run.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. If/Then Decisions: the case

02. Follow the deciding steps

03. Inclusive branching trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.